
Sage Plus Development and Benchmark Report

September 2026

ABSTRACT

Sage was developed by Michael R. Lazear and the upstream Sage contributors. Sage Plus is an independently versioned downstream distribution of that engine. Its extensions address input support, typed analytical output, memory use, execution controls, and modification-aware analysis. This report documents those changes and evaluates their consequences by comparing Sage v0.15.0-beta.2 with Sage Plus v0.1.0-beta.3 using matched public spectra, references, and search settings. The evaluation covered computational resources, worker scaling, identification agreement, peptide entrapment, synthetic phosphorylation, and label-free quantification. Repeated searches of selected human and mixed-species files used 30.1 and 21.2 percent less peak resident memory with Sage Plus. Runtime favored Sage Plus in the repeated public searches but favored Sage with entrapment-expanded references. Accepted peptide-spectrum match Jaccard indices ranged from 0.9661 to 0.9849. Disagreement included both changed assignments and identical assignments crossing different confidence thresholds. Mean peptide entrapment estimates were similar, with conditional difference intervals spanning zero. Matched quantification measured species-ratio accuracy, preparation variability, and coverage, while a human-only control revealed accepted foreign-species signal in both engines. Neither release produced a jointly accepted peptide set in the restricted synthetic phosphorylation challenge. The development achieved lower memory use while preserving substantial identification agreement on the tested workloads. Validation also exposed limits in peptide acceptance and recipient-file quantitative confidence, defining priorities for further development.

Keywords proteomics · database search · memory efficiency · entrapment · benchmarking · label-free quantification · phosphorylation

1 Introduction and motivation

This repository technical report documents changes and benchmarks maintained within the independent Sage Plus project. Sage was developed by Michael R. Lazear and the upstream Sage contributors¹. Credit for the original engine and its continuing maintenance belongs to that project. Sage Plus is an unofficial downstream distribution. This report does not imply authorship, review, or endorsement by Sage’s creators or maintainers.

Sage is an open-source engine for peptide identification and quantification from mass spectrometry data¹. Sage Plus is an independently maintained downstream distribution that extends this workflow with broader input support, structured analytical outputs, explicit

memory controls, and additional quantification and post-translational modification (PTM) functionality. The development problem was to make these capabilities available within a coherent search pipeline while managing their computational and scientific consequences.

These goals are related but require different forms of evidence. Compact storage can reduce the memory needed to represent a search database. New preprocessing and predictive models can change the evidence used to rank peptide candidates. Typed output can make downstream analysis easier to automate, while also exposing questions about what a confidence field means for a peptide, a modification site, or an intensity in one file. Successful execution alone resolves none of those scientific questions.

This report documents the development of Sage Plus and evaluates the behavior of Sage v0.15.0-beta.2 and Sage Plus v0.1.0-beta.3. It first describes the software changes and their design goals, then asks whether the resulting implementation altered resource use, identification behavior, error assessment, or downstream quantitative results. Accepted peptide-spectrum matches (PSMs) provide a regression endpoint for search behavior. Independent entrapment estimates, known mixture ratios, and a species-absent control extend the evaluation beyond agreement with upstream output.

The evidence combines frozen public pilot searches with a separately planned extension using the same released executables. Lower memory use and high PSM agreement were accompanied by workload-dependent runtime, foreign-species signal in a human-only quantitative control, and failed peptide acceptance in a synthetic PTM challenge. Compatibility failures and incomplete workloads are part of the development record because they identify boundaries that successful benchmarks alone would miss.

2 Development of Sage Plus

2.1 Relationship to upstream Sage and design goals

Sage Plus retains Sage’s search-engine foundation, Git history, authorship, MIT license, and citation metadata. Its independent version sequence distinguishes the downstream integration from releases maintained by the upstream project. The integration policy preserves upstream changes while allowing extensions to be developed and tested together. Sage’s indexed peptide search, target-decoy workflow, and identification-to-quantification pipeline provide the starting point for this work.

The development goals were to reduce large in-memory search structures, support additional data sources, make analytical results easier to consume reliably, and extend modification-aware analysis. Keeping these changes in one distribution also made their interactions part of the engineering problem. For example, retaining modification identity affects database expansion, fragment generation, quantification channels, and the exported site representation.

Table 1 summarizes the major changes relative to the evaluated Sage release. Motivation and expected consequence describe design intent. The later sections establish which consequences were observed. The inventory is anchored to the released source snapshots, with implementation paths in Section S2. It excludes subsequent working-tree repairs.

Area	Upstream Sage behavior	Sage Plus change	Motivation	Expected consequence
Input	mzML, MGF, and Bruker readers	mzMLb, Thermo RAW, peptide and cleavage inputs	Broader data and digestion workflows	Fewer conversions, new input contracts
Output	TSV and optional Parquet, wide LFQ tables	Typed Parquet, coordinates, long LFQ with nulls	Explicit identities and missing signals	Typed joins, revised consuming scripts
Storage	Peptide arrays and dense modification vectors	Protein spans and compact search structures	Reduce allocation and record size	Lower memory with possible access costs
Execution	CLI searching and file batching	Memory guards, API, events, isolated jobs	Bound resources and expose run state	Early rejection, inspectable failures
Modeling	Isotope processing and linear prediction	Scored envelopes, mass alignment, richer models	Represent candidate evidence more fully	Changed scores and acceptance thresholds
PTM	Mass-based static and variable modifications	PTM names, caps, neutral losses, site reports	Explicit hypotheses and site ambiguity	Bounded variants, new confidence endpoints
Quantification	LFQ tracing and reporter-ion quantification	Label channels, tracing controls, library export	Extend quantification and result reuse	More options, distinct validation needs

Table 1: Development areas in Sage Plus relative to Sage v0.15.0-beta.2. Changes describe the evaluated Sage Plus v0.1.0-beta.3 snapshot. Expected consequences are design hypotheses whose evaluation coverage is described in the text.

2.2 Input support and analytical output

Sage already supported mzML, Mascot Generic Format, Bruker data, and optional Parquet output. Sage Plus added local mzMLb and direct Thermo RAW readers, as well as pre-digested peptide input and additive protein-specific cleavage sites. These extensions support workflows whose data or digestion assumptions are difficult to express through the original input paths. Local seekable-file requirements remain part of the new reader interfaces. The matched searches in this report used common converted inputs, so they assess downstream behavior without isolating the fidelity or performance of each added reader.

Parquet became the canonical analytical representation in Sage Plus. Typed protein occurrences preserve sequence coordinates and flanking residues in PSM output. LFQ output uses one precursor and file per row, with missing integrated signals represented as nulls. This representation supports typed queries and explicit joins between search, quantification, and site results. It also changes the integration contract for existing scripts. The evaluation therefore reads the underlying result tables through a common normalizer before comparing identities or intensities.

2.3 Memory representation and execution controls

The memory work targeted structures that expand with the number of peptides, modification variants, theoretical fragments, and loaded spectra. Target peptide sequences can reference immutable source-protein storage by coordinates. Generated decoys and pre-digested peptides retain owned storage. Compact modification records share definitions across the database, and the theoretical fragment index stores lossless six-byte records with shared mass prefixes. Compact spectrum storage further reduces the resident representation. These changes aim to preserve sequence and mass information while reducing repeated allocations and per-record storage.

Representation changes can alter access costs even when they preserve stored information. Exact database prefiltering similarly trades additional preparation work for a smaller retained index.

Its paired target-decoy handling is part of the scientific contract as well as the implementation. Consequently, lower memory and shorter runtime were evaluated separately.

Memory estimation, runtime limits, minimum-free-memory protection, and configurable file batching address the operational problem of searches that outgrow a workstation. They provide early rejection or bounded execution when a requested search space is too large. The structured runner adds configuration validation, progress events, cancellation, and machine-readable run summaries. An optional Model Context Protocol (MCP) server manages persistent jobs in isolated worker processes. These are engineering interfaces. Their existence does not supply evidence about identification or quantification accuracy.

2.4 Scientific and modeling extensions

Other changes directly affect candidate evidence. Sage Plus uses average-scored isotope envelopes for tandem-spectrum preprocessing and carries fragment-charge confidence into matching. Per-file precursor and fragment mass-error alignment precedes final confidence rescoring. Retention-time prediction includes enriched features and regularized variable-PTM offsets with peptide-grouped folds. The mobility model also uses peptide-grouped prediction, and optional nonlinear alignment supports cross-run retention-time behavior. These mechanisms address isotope assignment, systematic mass error, and modification-dependent prediction. They can change score distributions and acceptance thresholds even when the main search workflow is retained.

The distinction between engineering and scientific changes therefore depends on what a change does to the evidence. Lossless fragment storage targets representation. Charge-aware fragment matching changes which observations can support a candidate. Output typing targets representation, while a new localization q-value introduces a confidence claim that requires an appropriate truth set and denominator. The evaluation compares complete executables, so attribution to individual mechanisms remains a question for controlled ablation experiments.

2.5 Quantification and modification functionality

Structured modifications add names, per-modification occurrence limits, total variant caps, and optional or required neutral-loss fragments. These controls make modification identity and search-space expansion explicit. A variant cap also changes which hypotheses are searched, so its effect belongs in the scientific interpretation of a configured workload.

Modification-defined precursor channels support labeled quantification through exact-mass partner extraction and reference ratios. LFQ gains configurable match-between-runs tolerance and an explicit switch for cross-run extraction. Empirical spectral-library export supports deterministic best-PSM selection or consensus spectra. The experimental library-search mode had been removed before the evaluated release. The present quantitative experiment tests unlabeled LFQ ratios, coverage, and a species-absent control. Labeled channels and library export are documented capabilities awaiting their own endpoint-specific evaluation.

PTM functionality includes localization, ambiguity-aware sequences, site reports, and target-decoy localization q-values. Its intended purpose is to carry modification evidence from peptide identification to interpretable site output. The synthetic challenge tests whether the released workflow can produce a jointly accepted peptide and site set. This places peptide acceptance

before site accuracy in the validation sequence and makes failure at either stage informative for further development.

3 Evaluation design

3.1 Software and comparison design

The Sage reference was commit `df9219951cc9a54cf4cd55d76541af24b687bd3d`. Sage Plus was the released beta.3 executable associated with tag `v0.1.0-beta.3`. Executable hashes, source identity, configurations, and input receipts were frozen before analysis. Section S1 records the identities required to reproduce the comparison.

Each matched pair used the same spectra, reference database, search settings, and worker count. Searches ran sequentially on an Intel Core i7-10700K system under Linux. The principal comparisons used eight Rayon workers. A common process address-space ceiling of 16 GiB bounded allocation. Sage Plus also applied its configured 10 GiB memory guard. The two controls measure different resource limits.

The experiment compares complete executable behavior under matched inputs. Identical configuration keys can activate different implementations, as described in Section 2. Default modeling behavior was retained within each frozen executable.

3.2 Public spectra and reference databases

The human embryonic kidney (HEK) dataset, Proteomics Identifications Database (PRIDE) accession PXD001468, supplied two fraction acquisitions from a published proteomics study². Tandem mass spectra were extracted from mzXML into Mascot Generic Format without additional centroiding. Charge, precursor, retention time, and array lengths were checked during conversion. This derivative contains tandem spectra for identification and cannot support an evaluation of precursor quantification.

PRIDE accession PXD028735 supplied four Orbitrap data-dependent acquisition files from the Alpha and Beta preparations of mixture conditions A and B³. The first injection from each preparation was selected from metadata. ThermoRawFileParser `v.2.0.0-dev` converted the vendor files to mzML with vendor peak picking and both precursor and tandem spectra retained. Both engines received identical converted files. Preparation labels distinguish the Alpha and Beta preparations.

Reference proteomes came from UniProt release `2026_03`. The human search used reviewed canonical human proteins. The mixture used human, *Saccharomyces cerevisiae*, and *Escherichia coli* proteins with the stated retention-time standards. The yeast identity followed the primary methods and reagent documentation. Reference receipts preserve the selected accessions and downloaded contents.

An explicit amendment was required for the mixture reference. Sage Plus rejected entries containing undefined X residues. The amended reference excluded the affected proteins in their entirety for both engines, including their otherwise defined subsequences. The original searches, reference, and rejection records were preserved. All completed mixture comparisons reported here use this same amended reference. The reference amendment defines the effective mixture search space.

3.3 Search settings and identification agreement

The public search used tryptic specificity, one missed cleavage, and peptide lengths from 7 to 50 residues. Carbamidomethylation of cysteine was fixed and variable modifications were absent. Precursor and fragment tolerances were respectively ± 10 and ± 20 parts per million. The search reported one PSM per spectrum and disabled chimeric assignments. Full settings, including isotope errors and fragment charge limits, remain in the frozen configurations.

A common extractor read underlying Sage tabular results and Sage Plus Parquet results. It selected target PSMs at the reported 1 percent spectrum q-value threshold. Agreement used matched spectrum and peptide identities rather than release-specific run-summary counters. The Jaccard index is the intersection of the accepted PSM sets divided by their union. Engine-specific discoveries were retained separately.

3.4 Repeated timing and memory

One metadata-selected file from each public study entered the repeat timing experiment. Each engine received one warmup and three measured trials, with engine order alternating across trials. Timing began after acquisition, conversion, and the primary search matrices had finished. GNU Time recorded whole-process wall time and maximum resident set size. Reported medians exclude warmups. Observed minimum and maximum times describe repeat variability on this host, not confidence intervals across biological samples.

Wall time includes database construction, reading spectra, search, modeling, and writing output. It excludes the preceding acquisition and conversion. Peak resident memory differs from the address-space allocation limit. The two public studies use different input formats and workloads, so absolute timings should be compared within each study. A separate local HEK comparison is retained in the Supporting Information as a contextual observation because it could overlap acquisition activity.

3.5 Independent peptide error assessment

Entrapment searches used the human reference and the amended mixture reference, with two selected files per study and three shared database-construction seeds. Each constructed database was searched by both engines. FDRBench 1.1.1 supplied the paired shuffled entrapment estimator⁴. Sequence extraction and digestion used the documented reference rather than a small protein subset.

The independent audit retained one actual best-discriminant PSM per peptidoform and checked consistent peptide q-values. It reproduced target and entrapment counts at nominal peptide q-values of 0.1, 0.5, 1, 2, and 5 percent. These static-modification searches required one peptidoform per sequence. The paired estimator was bounded across exact target-partner score ties, and the official FDRBench value was checked against those bounds. The conservative bound supplies the reported comparison. Empty denominators remain undefined.

A false discovery proportion (FDP) describes a realized discovery set. False discovery rate (FDR) concerns an expectation over repetitions of an experiment. This report reports entrapment estimates as FDP.

Study summaries averaged the file-by-seed estimates. A bootstrap with 10,000 resamples kept the engines paired and resampled files and shared construction seeds jointly across engines. The

percentile interval describes uncertainty conditional on the selected files and construction seeds. Individual spectra and peptides were not treated as independent experimental replicates.

3.6 Analysis of identification differences

We distinguished acceptance differences from changes in the underlying spectrum assignment. An engine-only accepted PSM was first looked up in the complete rank-one output of the other engine. An identical spectrum, charge, and peptide assignment above that engine’s spectrum q-value threshold was classified as a threshold difference. A different rank-one assignment for the same spectrum and charge formed a separate category. The remaining category contained unmatched spectrum and charge combinations. A different charge assignment can therefore enter the unmatched category. These categories describe where the exported results diverge.

Threshold-response curves used the same files at every operating point. The relative change was the Sage Plus target count divided by the Sage target count, minus one. We computed that change separately for spectrum matches and peptidoforms. Curves summarize file-level changes within each study, with the observed file range retained.

3.7 Matched worker-scaling extension

The additional worker experiment used the previously selected first HEK input. Each released engine ran at 1, 2, 4, and 8 Rayon workers. Every engine and worker-count combination had one warmup followed by three measured searches. Engine order alternated across trials within a worker count. The common search configuration and executable identities were fixed in a new plan before execution. The memory guard settings remained those used in the original comparison.

Whole-process time and resident memory were collected with the same runner as the original pilot. Speedup was the median one-worker time divided by the median time at the specified worker count. It includes serial work and input/output, so it measures application scaling rather than the parallel efficiency of the fragment-matching kernel alone. The workstation was not isolated from its normal background services. We report the extension separately from the earlier public timing experiment and retain every warmup and measured result.

3.8 Synthetic phosphorylation challenge

The phosphorylation challenge evaluated a PTM search setting. It used higher-energy collisional dissociation (HCD) acquisitions from PRIDE PXD000138 and synthesis-defined peptide sequences⁵. The restricted database contained the synthetic sequences, with one variable phosphorylation permitted on serine, threonine, or tyrosine. Peptides were searched as intact library entries, with fixed cysteine carbamidomethylation and a shared target-decoy configuration. This restricted search does not reproduce the original full-background database experiment.

Both releases were evaluated at the spectrum-only threshold and at the joint spectrum and peptide thresholds. The primary localization endpoint additionally required an accepted site with a reported localization q-value. Sage Plus exports site-level results, but the Sage output used here does not provide an identical site-confidence field. We therefore compared the shared upstream acceptance stages first. A failure to obtain accepted peptide identifications precludes an interpretable comparison of accepted site accuracy.

Synthesis consistency was retained as a secondary Sage Plus diagnostic in the Supporting Information. It pools the available library truth because the mapping between acquisition files and individual synthetic libraries was not independently verified. It combines identification and localization discrepancies and cannot be interpreted as arrangement-level false localization rate. Truth-derived site-prior experiments were not used to claim predictive performance.

3.9 Matched label-free quantification

The quantification extension jointly searched the four mixture acquisitions and the human-only control from PXD028735. Both releases used the same amended reference, search settings, batch size, and eight-worker setting. Shared label-free quantification (LFQ) settings specified hybrid peak scoring, summed integration, a spectral-angle cutoff of 0.7, a 5 parts-per-million mass tolerance, combined charge states, and a 1 percent peptide threshold. Default cross-run extraction behavior was retained in both releases. An engine-specific match-between-runs toggle was not introduced into this matched comparison.

Sage’s wide LFQ table was converted to one precursor and file per row. Sage Plus’s long Parquet table was normalized to the same representation. Combined charge states were mapped to a common empty charge key. The Sage export omits decoy quantification rows, so only target rows entered comparative yield and ratio endpoints. Protein inference was not assumed identical between engines. Species assignment required an unambiguous protein-species mapping. Sequences occurring in more than one reference proteome after isoleucine/leucine normalization were excluded from mixture ratio analysis.

Primary quantitative results used positive finite intensities at a precursor LFQ q-value of at most 1 percent. The 5 percent threshold was reported separately to expose threshold-dependent yield. For each preparation, we computed log-base-two B/A ratios without imputation. Expected values were 0 for human, -1 for yeast, and 2 for *E. coli*, following the mixture design³. Bias was the median difference from the expected ratio. Absolute error was the median absolute difference. Preparation variability was the coefficient of variation (CV) across the Alpha and Beta intensities within a condition. These pairs measure preparation variability.

Missingness used the union of features observed in at least one of the four mixture files for that engine. The denominator was that feature count multiplied by the number of mixture files. It does not include all theoretically detectable peptides. To separate changes in selected feature sets from changes in their intensities, we also compared ratio error on identical peptide and preparation pairs accepted by both releases.

The human-only control supplied a species-absent diagnostic. A foreign assignment had to map only to yeast or *E. coli* proteins and lack an isoleucine/leucine-equivalent occurrence in the human reference. We counted foreign positive rows both in the complete accepted control set and among rows without direct accepted tandem spectral evidence. Direct tandem mass spectrometry (MS2) evidence required the same peptide in the same file at both spectrum and peptide q-values of at most 1 percent. This common rule replaced release-specific confirmation flags. A precursor q-value shared across files was not treated as an individual recipient-file transfer q-value.

4 Computational performance

4.1 Public search time and memory

Sage Plus used less peak resident memory on both repeated public workloads. For the HEK file, median memory was 1491.8 MiB for Sage Plus and 2134.1 MiB for Sage. For the mixture file, the corresponding values were 2419.9 and 3071.8 MiB. These are reductions of 30.1 and 21.2 percent relative to Sage.

Median wall time was also lower for Sage Plus in these two workloads. HEK searches took 9.48 seconds with Sage Plus and 9.91 seconds with Sage. Mixture searches took 25.03 and 26.05 seconds, respectively. The reductions were 4.3 and 3.9 percent. Figure 1 summarizes the measured trials, and Table 2 records medians and observed ranges.

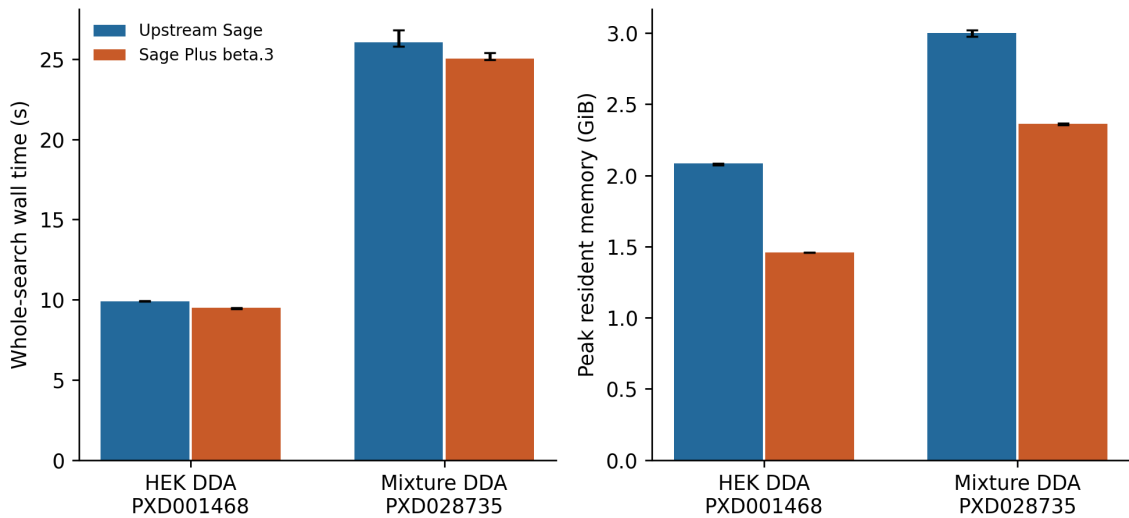


Figure 1: Repeated public searches comparing Sage v0.15.0-beta.2 and Sage Plus v0.1.0-beta.3. One selected file per study was searched with one warmup and three measured trials per engine at eight workers. Bars show medians and error bars span observed minima and maxima. Timing followed acquisition and conversion. These repeats characterize the selected workloads on one host.

Study	Engine	Seconds (range)	Peak MiB	Target PSMs
PXD001468	Sage	9.91 (9.91–9.94)	2134.1	16,866
PXD001468	Sage Plus	9.48 (9.43–9.50)	1491.8	16,917
PXD028735	Sage	26.05 (25.80–26.79)	3071.8	55,457
PXD028735	Sage Plus	25.03 (24.95–25.38)	2419.9	55,612

Table 2: Public timing medians for the specified Sage and Sage Plus releases. Parentheses contain observed minimum and maximum wall times, not confidence intervals. Peak memory is maximum resident set size in MiB. Target PSMs pass the reported one-percent spectrum q-value threshold. Warmups are excluded.

The runtime direction changed when the database was enlarged for entrapment. Sage Plus took longer while retaining lower median peak memory in both studies. Table S1 reports these measurements separately because they span files and construction seeds rather than repeat trials of one fixed input. The local conventional and common-modification searches also favored

Sage on runtime, as shown in Table S2. Together, these observations establish a workload-dependent runtime tradeoff.

Figure 2 places these resource results on a common relative scale. The memory advantage persisted across public, entrapment, and local modified-search workloads, while the time ratio crossed the equal-performance reference. In the entrapment comparisons, the median time changes were +19.1 percent for HEK and +18.5 percent for the mixture. Their corresponding memory changes were -18.6 and -18.6 percent. The larger search space exposed a time-memory tradeoff that the conventional public timing alone would conceal.

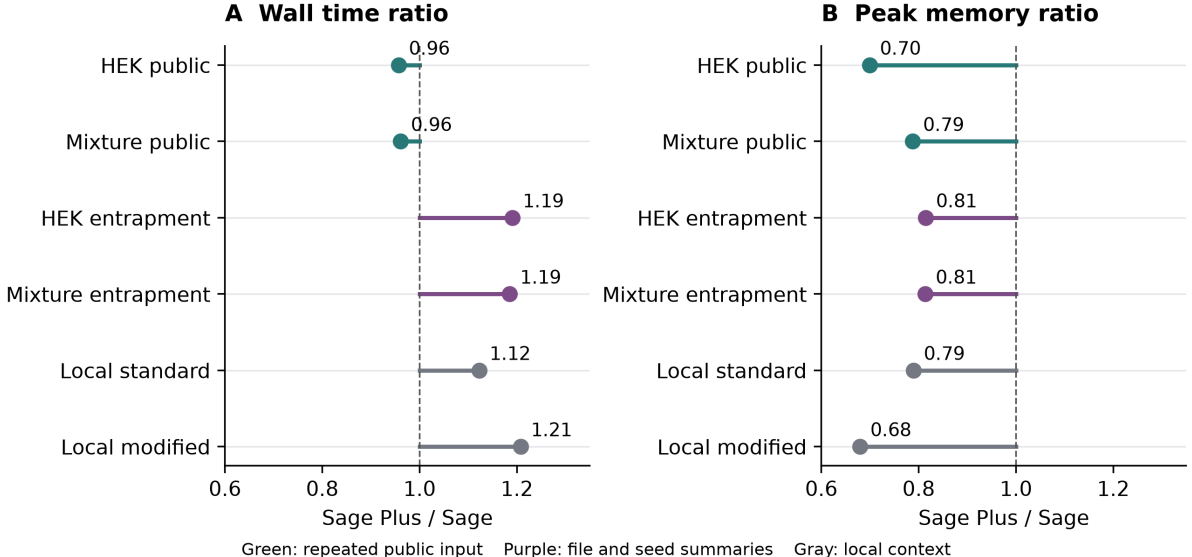


Figure 2: Within-workload Sage Plus to Sage ratios for wall time (A) and peak resident memory (B). Values below one favor Sage Plus. Public points summarize repeated searches of a fixed input. Entrapment points summarize different files and database-construction seeds. Local points are contextual because acquisition activity could overlap them. Connecting segments are visual references to equality, not uncertainty intervals.

4.2 Worker scaling

Both releases benefited from additional workers on the selected HEK workload. The observed one-to-eight-worker speedups were 3.76 for Sage and 4.06 for Sage Plus. Their eight-worker median times in this extension were 10.42 and 9.82 seconds, respectively. Figure 3 shows the full response and individual measured repeats.

The scaling experiment shows that the lower-memory implementation retains useful parallel execution on this workload. Wall time contains work that cannot scale linearly with workers, including reading the input and writing results. Peak resident memory is also a whole-process maximum, not a per-worker allocation. The recorded endpoint therefore informs deployment on this workstation while leaving other core counts, architectures, and concurrent batch workloads unmeasured.

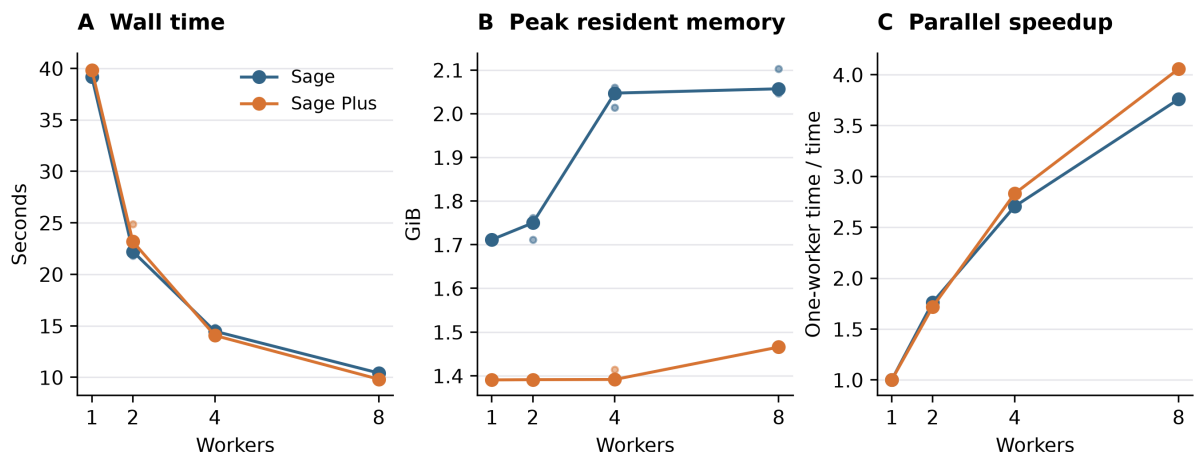


Figure 3: Matched worker scaling for the frozen release pair on the first HEK input. Panels show wall time (A), peak resident memory (B), and speedup relative to each engine’s one-worker median (C). Lines join medians of three measured searches after one warmup per engine and worker count. Small points show individual measured searches. Warmups are excluded. Engine order alternated within each worker count.

5 Identification behavior and error assessment

5.1 Agreement of accepted identifications

The completed public comparison contained 6 matched file pairs. PSM Jaccard indices ranged from 0.9661 to 0.9849. Both engines reported discoveries absent from the other, despite the high overlap. Table 3 shows the shared and engine-specific counts for every completed pair.

Input pair	Shared PSMs	Sage only	Plus only	Jaccard
PXD001468-0	16752	114	165	0.9836
PXD001468-1	19291	115	180	0.9849
PXD028735-0	54639	818	973	0.9683
PXD028735-1	55090	850	954	0.9683
PXD028735-2	57416	921	1092	0.9661
PXD028735-3	56282	892	1039	0.9668

Table 3: Accepted target PSM agreement at each engine’s reported one-percent spectrum q-value threshold. File indices identify the frozen input order in Section S3. Mixture rows use the amended reference for both engines. Jaccard is the shared count divided by the union of accepted PSM identities.

Sage Plus reported slightly more accepted PSMs and peptidoforms in every public file at the primary threshold. Across the HEK files, the per-file PSM changes ranged from +0.30 to +0.33 percent. Across the mixture files, they ranged from +0.19 to +0.29 percent. The corresponding peptidoform changes were +0.17 to +0.31 percent for HEK and +0.15 to +0.32 percent for the mixture.

Figure 4 shows both identification levels and their threshold response. The magnitudes are small relative to the total discovery sets. The range across files is retained because averaging

the counts could hide a file whose direction differed from the study mean. The threshold curves also show that an engine's relative yield is a property of the selected operating point.

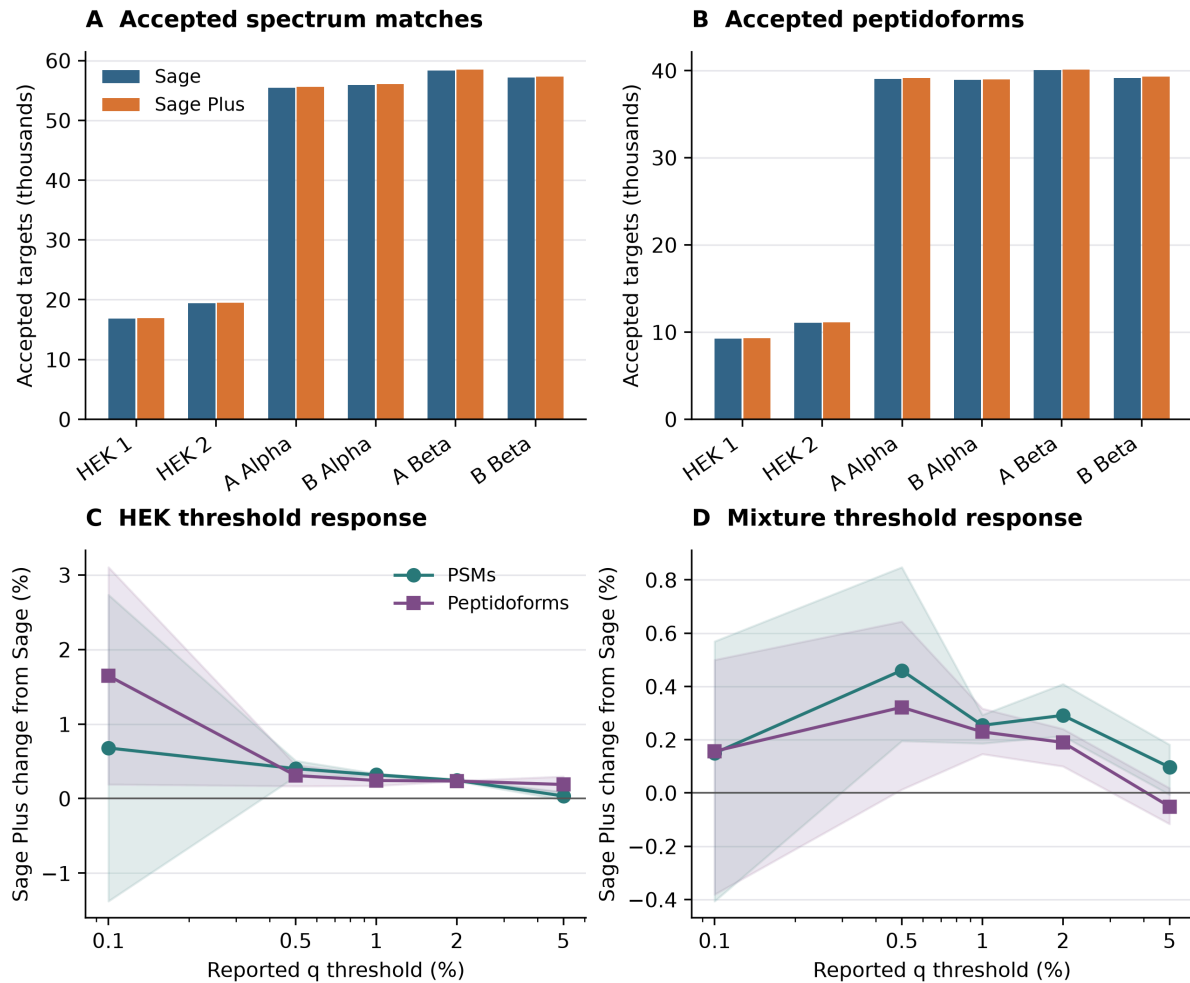


Figure 4: Identification yields from matched public files. Panels A and B show target PSM and peptidoform counts at the one-percent spectrum and peptide thresholds, respectively. Panels C and D show the mean file-level percentage change from Sage to Sage Plus across nominal thresholds. Shading spans the observed file range, not a confidence interval. HEK labels follow the input order in the Supporting Information. Mixture labels identify condition and preparation. All comparisons use matched amended references where required.

The disagreement analysis showed that an accepted PSM unique to one engine was not always a different peptide assignment. Among Sage-only accepted PSMs, 39.5 percent had the identical assignment in Sage Plus above its acceptance threshold. The reciprocal fraction among Sage Plus-only PSMs was 44.6 percent. Figure 5 separates these threshold crossings from changed rank-one assignments and unmatched spectrum-charge combinations.

Changed assignments remain relevant even when aggregate yields are close. Agreement at the level of total counts would miss such substitutions. Conversely, an engine-only count should not be read as the number of spectra the other engine failed to search. The decomposition identifies which disagreements could be revisited by an operating-point analysis and which require inspection of competing spectrum assignments.

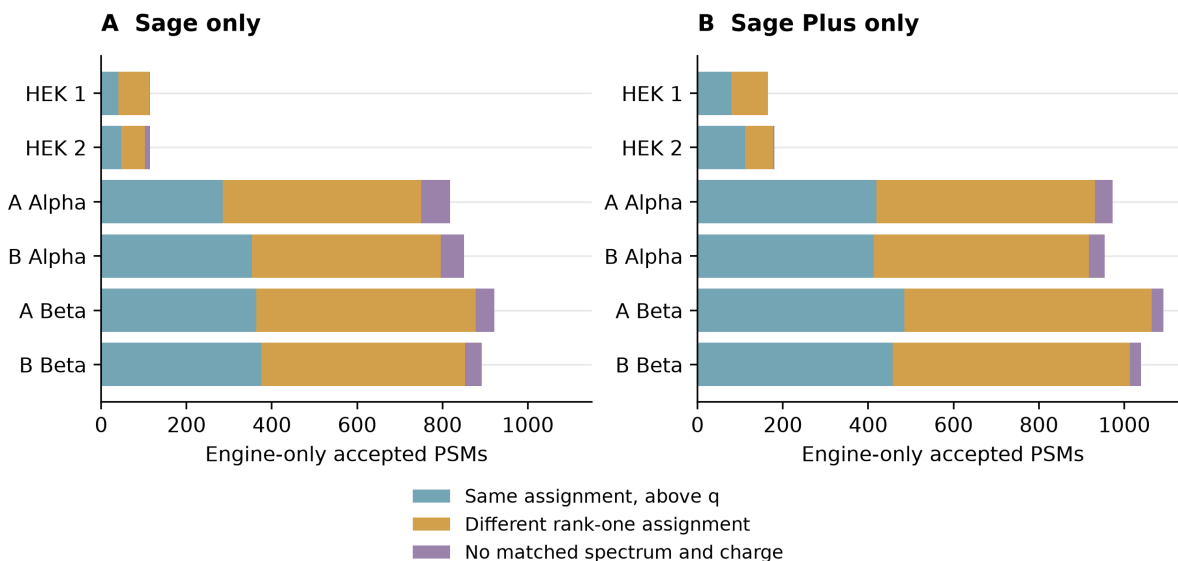


Figure 5: Origin of accepted PSM disagreement at the one-percent spectrum threshold. Sage-only assignments appear in A and Sage Plus-only assignments in B. Categories indicate an identical assignment above threshold in the other engine, a different rank-one assignment at matched spectrum and charge, or no matching rank-one spectrum and charge. The last category includes possible charge changes. Counts are descriptive, and no category establishes which peptide assignment is correct.

Acceptance depends on each engine’s score distribution and estimated q-values. The next analysis examines the independently estimated error behavior of the peptide discovery sets.

5.2 Peptide entrapment estimates

At the nominal 1 percent peptide q-value threshold, mean paired FDP for the HEK study was 1.017 percent for Sage and 0.998 percent for Sage Plus. For the mixture study, the estimates were 1.029 and 1.057 percent, respectively. The engines therefore gave similar mean estimates in this selected pilot, with the direction of the difference varying by study.

The Sage Plus minus Sage difference was -0.019 percentage points for HEK, with a descriptive interval from -0.083 to +0.054. For the mixture, it was +0.028 percentage points, with an interval from -0.039 to +0.066. Both intervals include zero, leaving the direction of the difference unresolved in this pilot. Figure 6 shows the threshold profiles, and Table 4 reports the nominal one-percent comparison.

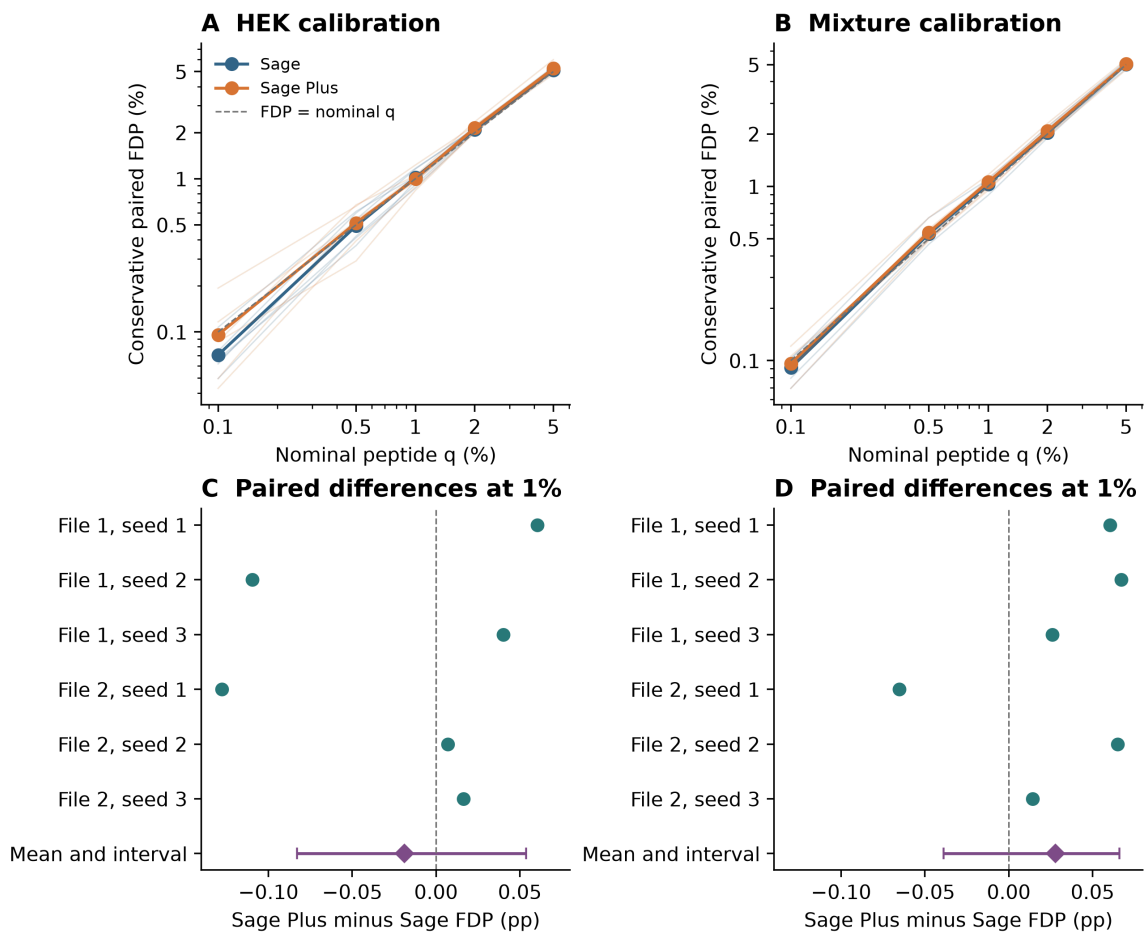


Figure 6: Independent entrapment results for the frozen Sage and Sage Plus release pair. Panels A and B show mean conservative paired FDP profiles, with faint lines for individual files and construction seeds. Panels C and D show each file-by-seed difference at the primary threshold and the study mean with its conditional bootstrap interval. Seed labels follow the order in the Supporting Information. The nominal peptide q-value is distinct from independently estimated FDP. Reference construction and tie handling are described in Section 3.

Study	Sage FDP (%)	Plus FDP (%)	Difference (pp)	Interval (pp)
HEK	1.017	0.998	-0.019	-0.083 to +0.054
Mixture	1.029	1.057	+0.028	-0.039 to +0.066

Table 4: Mean paired FDP at the nominal one-percent peptide threshold. Difference is Sage Plus minus Sage in percentage points (pp). Intervals are the conditional bootstrap percentile bounds across selected files and shared construction seeds.

The threshold-specific target counts and mean FDP estimates appear in Table S3. They expose the dependence on the chosen operating point. Reference coverage, entrapment construction, and the relationship between target and entrapment peptides remain assumptions of the estimator⁴.

6 Downstream analytical behavior

6.1 Synthetic PTM acceptance

Evaluation of the new PTM functionality identified a limit of the released workflow. The restricted phosphorylation challenge produced the same spectrum-level accepted counts in both releases. Each engine accepted 3,141 target PSMs in the first HCD file and 1,958 in the second. Requiring the peptide threshold reduced the accepted sets to 0 for both releases and both files. Figure 7 shows this loss at the confidence-filtering stage.

All retained rank-one matches in these runs had peptide q-values equal to one. Thus the absence of primary accepted sites did not arise solely from a Sage Plus localization cutoff. It was already present at the shared peptide acceptance stage. The diagnostic site output should therefore not be promoted to a validated phosphosite set by omitting the peptide filter. The restricted synthetic database and sparse decoy evidence also distinguish this challenge from a broad phosphoproteome search.

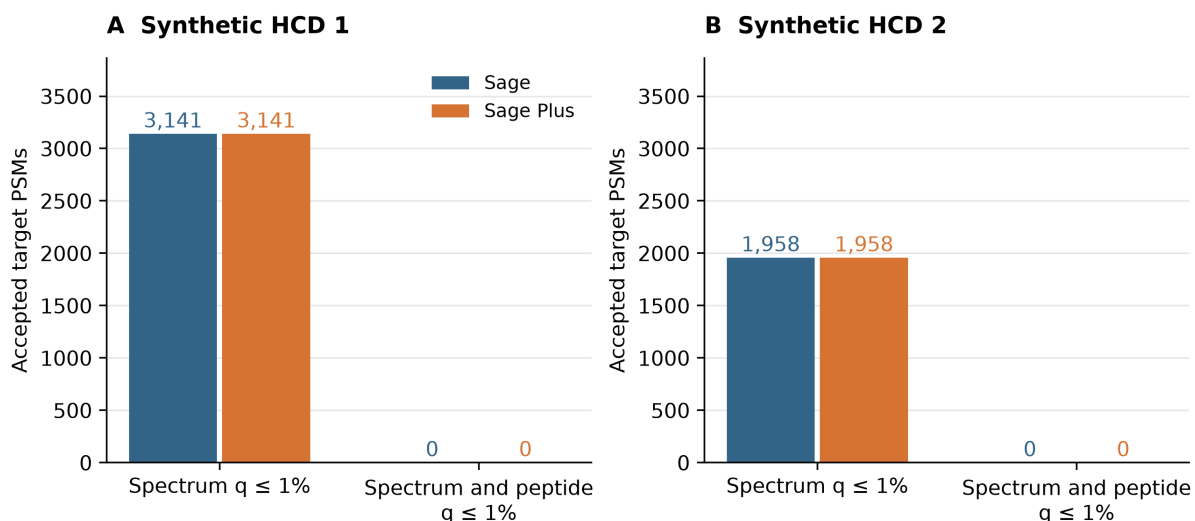


Figure 7: Acceptance stages in the matched synthetic phosphorylation search. Panels A and B correspond to the two HCD input files. Bars compare target PSMs passing the spectrum threshold alone with those passing both spectrum and peptide thresholds at one percent. The joint accepted sets are empty in both engines. The figure reports successful search outputs with failed primary acceptance, not search failures or a zero localization-error estimate.

6.2 Matched quantitative accuracy and coverage

The matched LFQ extension completed in both releases. At the primary precursor threshold, Sage retained 25,018 target precursors and Sage Plus retained 24,510. Their positive target precursor-file counts were 123,687 and 121,179, respectively. These counts span the jointly searched mixture and control files. They are not protein counts or counts of independently confirmed recipient-file transfers. Sage Plus precursor yield changed by -2.03 percent at the primary threshold and by +0.24 percent at the secondary threshold. The direction therefore depended on the confidence cutoff.

Figure 8 displays the species-specific ratio distributions, absolute ratio errors, preparation CVs, and missingness. Each ratio uses a matched preparation pair with positive intensity in both

conditions. The expected species shifts provide a stronger quantitative check than a correlation between engine intensities alone. A high correlation could persist despite a common ratio bias or many incorrect transfers.

Human log-ratio biases were -0.044 for Sage and -0.045 for Sage Plus. Yeast biases were -0.142 and -0.141, while *E. coli* biases were +0.183 and +0.189. Table 5 records the ratio denominators and error summaries for each species. The histograms expose the distribution around those summaries rather than treating a median as a complete accuracy result.

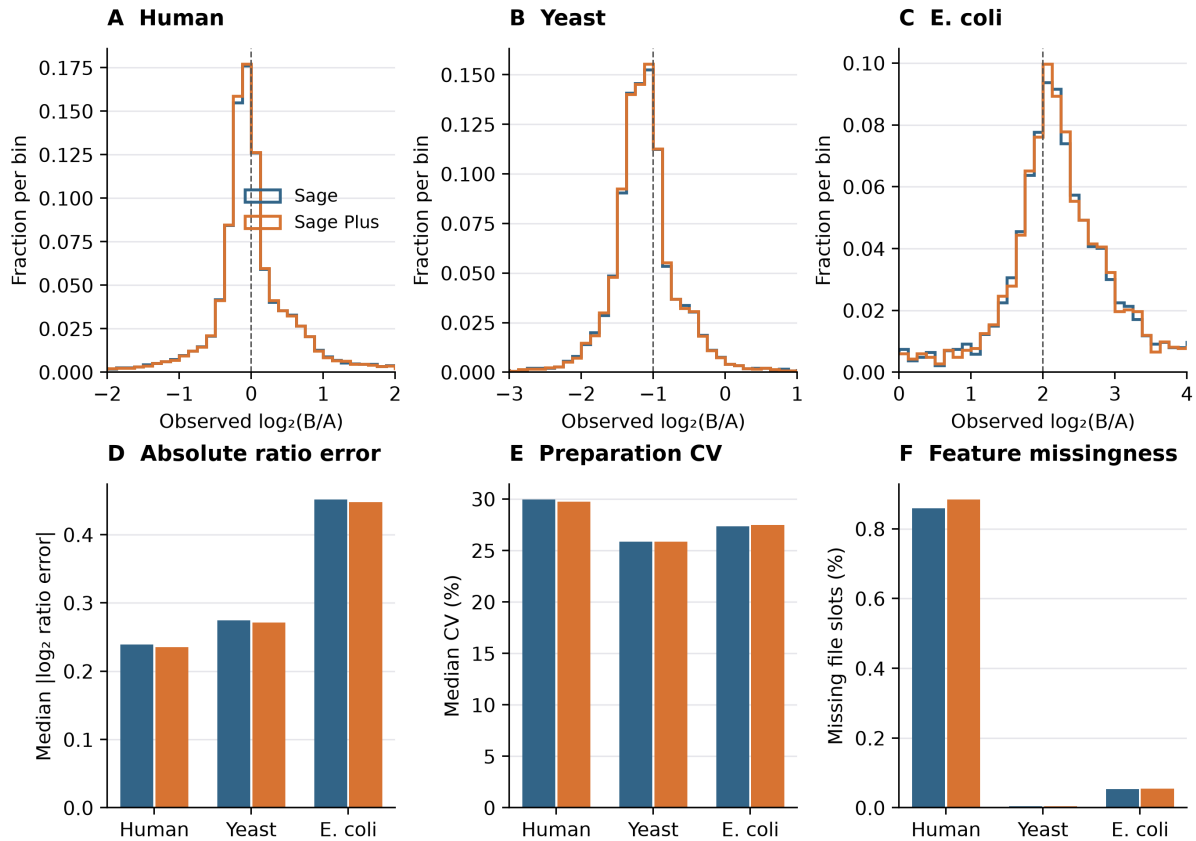


Figure 8: Matched LFQ performance at one-percent precursor q-value. Panels A through C show per-preparation log-base-two B/A ratios for human, yeast, and *E. coli*. Dashed lines mark the expected ratios. Histograms use equal-width bins normalized by each engine's full ratio count, with tails outside the displayed interval omitted visually. Panels D through F show median absolute ratio error, median preparation CV, and missingness over each engine's observed feature union. No intensity was imputed. Features with cross-species isoleucine/leucine ambiguity were excluded.

Engine	Species	Ratio pairs	Bias	Abs. error	CV (%)	Missing (%)
Sage	human	35,975	-0.044	0.239	29.9	0.859
Sage	yeast	10,583	-0.142	0.275	25.9	0.005
Sage	ecoli	1,866	+0.183	0.451	27.4	0.054
Sage Plus	human	35,230	-0.045	0.235	29.7	0.884
Sage Plus	yeast	10,439	-0.141	0.272	25.9	0.005
Sage Plus	ecoli	1,826	+0.189	0.447	27.5	0.055

Table 5: Quantification endpoints for the matched released engines at the primary precursor threshold. Ratio pairs combine a peptide and preparation. Bias and absolute error are on the log-base-two B/A scale. CV describes Alpha versus Beta preparation variability within condition.

Missingness uses the engine-specific observed union, not the theoretical proteome.

The engines’ accepted quantitative sets need not contain the same precursors. The shared-set analysis retained 33,792 human, 10,183 yeast, and 1,788 *E. coli* peptide-preparation ratio pairs. Table S6 compares absolute error on these identical pairs. This check separates intensity agreement on common features from the effect of selecting different quantitative sets. Across species, the median absolute difference between engine log-ratios on common pairs ranged from 0.0062 to 0.0067. This close agreement shows stable quantitative behavior on the shared features.

6.3 Species-absent control results

The human-only control contained accepted positive intensities assigned to foreign species in both engines. Sage reported 5,965 foreign rows among 24,262 assessable quantified rows. Sage Plus reported 5,872 among 23,771. These assignments survived exclusion of foreign sequences also found in the human reference after isoleucine/leucine normalization.

Among control rows without direct jointly accepted spectral evidence, foreign assignments accounted for 66.0 percent in Sage and 66.6 percent in Sage Plus. Figure 9 displays the counts and the dependence of overall precursor yield on the LFQ threshold. Table S7 retains the exact numerators and denominators used for the control calculation.

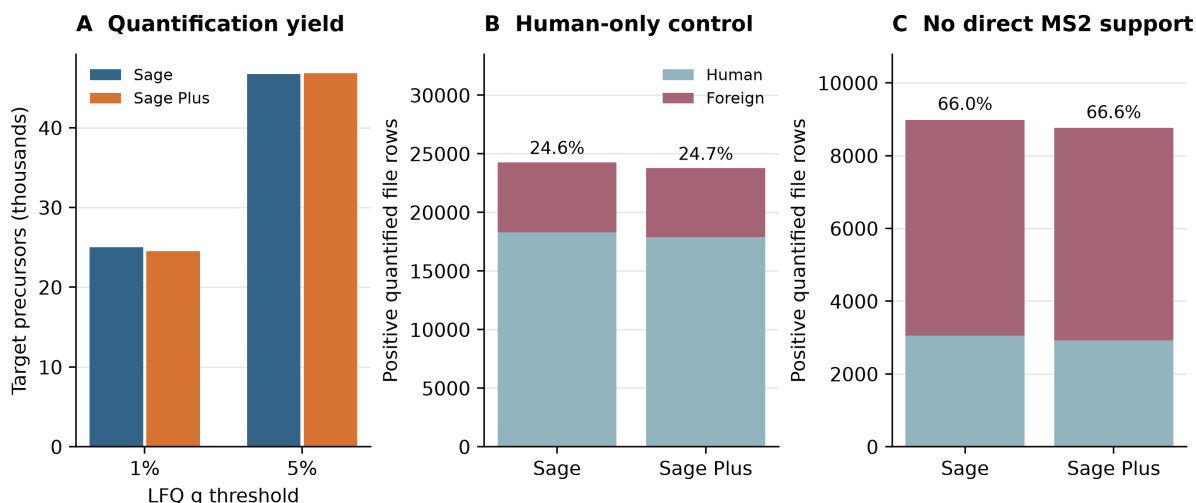


Figure 9: LFQ yield and human-only control diagnostics. Panel A shows target precursor counts at one- and five-percent LFQ q-values across all searched files. Panel B partitions assessable positive control rows at the primary threshold into human and foreign assignments. Panel C applies the same partition after excluding rows with direct spectrum and peptide evidence at one percent. Labels above bars give foreign fractions. Fractions describe the indicated control subsets.

The absent-species control exposed a limitation in treating a shared precursor-level confidence score as evidence for an intensity in each file. Foreign signal persisted even among rows lacking direct accepted MS2 evidence. This result motivates explicit recipient-file confidence assessment alongside mixture-ratio accuracy.

6.4 Compatibility and incomplete workloads

The undefined-residue reference rejection was a compatibility difference with a practical effect on reproducibility. Completing the mixture comparison required a documented common reference amendment. Reporting only the amended successful runs would conceal this limitation, so the original failures remain part of the study record.

Neither engine completed the broad modification stress workload under the selected resource limits. Sage Plus stopped at its modified-database memory guard, while Sage failed allocation under the process ceiling. A guard failure and an allocation failure are distinct outcomes. Neither supports a completed-search time, accepted identification count, or relative performance estimate. These observations bound the workloads covered by the successful comparison.

7 Engineering and scientific lessons

7.1 Memory reduction changes the performance question

The development produced a consistent memory benefit across the completed workloads, with a runtime tradeoff that depended on search-space conditions. Compact internal representation is a plausible contributor to this pattern: less storage can require different access, reconstruction, or preparation work. The complete-executable comparison does not isolate that mechanism. The useful engineering lesson is to evaluate capacity and throughput separately. A smaller resident search can make a workload practical even when execution takes longer. Conversely,

a faster conventional search does not predict runtime for a larger modified or entrapment-expanded database.

Resource guards also make failure behavior part of the interface. Early rejection at a database guard and allocation failure at a process ceiling both prevent completion, but they provide different information for the next run. Retaining those outcomes documents what must change in the search space or resource budget before a completed-search comparison becomes possible.

7.2 Regression evidence requires identities and confidence stages

Major implementation changes preserved high identification agreement without producing identical accepted result sets. This makes PSM identity overlap a useful regression measurement. Aggregate yield alone would miss substitutions between peptide assignments. The disagreement decomposition adds another layer: an assignment can be present in both outputs and accepted by only one engine because its confidence score crosses a threshold.

Threshold crossings direct development toward score distributions and confidence estimation, while changed assignments direct attention toward preprocessing, candidate generation, and ranking. Independent entrapment then checks a different property of the accepted sets. Similar observed entrapment behavior provides evidence that the changes preserved error behavior in these workloads, subject to the estimator’s reference assumptions. Shared ancestry makes this independent check particularly useful because agreement between implementations can preserve common limitations.

7.3 Quantitative confidence must match the reported unit

The LFQ evaluation showed that close agreement on shared mixture features can coexist with foreign-species signal in a human-only file. During validation of the expanded quantitative functionality, this absent-species control exposed a limit in interpreting precursor-level confidence as file-specific evidence. The design lesson is to distinguish evidence that a precursor was identified somewhere in a joint analysis from evidence for its intensity in a particular recipient file.

Long-form output makes that distinction easier to inspect, but a row and a confidence field do not by themselves resolve it. Quantitative coverage, known ratio accuracy, and control behavior need separate endpoints. Future output and model development should make their units explicit so that consumers can decide which intensities satisfy the evidence requirements of a downstream analysis.

7.4 Exported functionality requires end-to-end validation

The PTM challenge located a failure before site acceptance: spectrum matches were available, but the peptide confidence requirement removed the primary set. Adding a localization field therefore did not establish usable localization performance on this challenge. Stage-wise validation made the empty result actionable by identifying the first acceptance barrier. Future work should verify an accepted peptide set on independently mapped synthetic libraries before interpreting accepted-site accuracy.

The undefined-X reference rejection provides a related compatibility lesson. Input validation determines which biological hypotheses reach the search. A common amended reference enabled comparison, while retaining the original failure preserved the cost of that accommodation.

Together, the reference, resource, and PTM outcomes show why failed workloads belong in a permanent technical record of software development.

8 Limitations and future development

The core identification comparison covers selected files from two public studies and one host. Timing repeats characterize repeated computation, while the synthetic phosphorylation challenge adds a restricted library setting. Preparation pairs and many precursor measurements do not increase the number of independent biological studies. The extension reused previously inspected inputs and is post hoc. The evidence therefore characterizes these released workflows and does not establish universal superiority or held-out generalization.

Confidence comparisons also have defined boundaries. Nominal q-value thresholds do not impose equal independently estimated error. The entrapment intervals are conditional on the selected files and construction seeds, and no power analysis or equivalence margin was specified. They cannot establish equivalence or general FDR calibration. The species-absent control cannot detect erroneous human-to-human transfers, and contamination or reference errors can contribute foreign signal. Its fractions are control diagnostics rather than a general LFQ transfer-error rate. Empty jointly accepted PTM sets leave accepted-site accuracy unresolved for this challenge.

The amended mixture reference omits entire proteins with undefined residues. Human isoform coverage, strain identity, and incidental contaminants can also affect the effective search space. Conversion software and executable build choices are additional parts of the tested workflow. Common converted inputs leave the new native readers unevaluated, while labeled channels, empirical library export, and the automation interfaces require dedicated validation.

The released pipeline fits base retention-time and final discriminant models using observations that also receive scores. Grouped folds for an additive PTM component do not make the whole pipeline independent of its evaluation data. Held-out model evaluation should address this reuse. Controlled ablations should separate the effects of compact storage, preprocessing, prediction, and execution controls under matched resource budgets.

Future evaluation should freeze software identities, references, extraction rules, and primary endpoints on reserved files and additional independent studies. Priorities are identification yield at independently matched error, jointly accepted PTM site accuracy, and recipient-file quantitative confidence. Each requires explicit denominators and a sample design appropriate to the intended claim. Repairs developed after the evaluated snapshot need new matched runs with their own software identity.

9 Conclusion

Sage Plus extends Sage through changes to input support, analytical output, memory representation, execution controls, and modification-aware analysis. Evaluating those changes established a practical memory benefit and high identification agreement on the tested workloads, together with workload-dependent runtime and similar observed peptide entrapment behavior. Downstream validation exposed limits in recipient-file quantitative confidence and synthetic PTM peptide acceptance. Compatibility and resource failures further defined the implementation’s operating boundaries. These results form a development record of what

remained stable, what improved, and what requires further work in the released Sage Plus workflow.

Associated Content

Supporting Information

The appended Supporting Information contains software identities, input mapping, reference amendments, paired resource measurements, identification counts, entrapment tables, quantification denominators, synthetic-site diagnostics, and reproducibility commands.

Attribution and report status

The original Sage engine is the work of Michael R. Lazear and the upstream Sage contributors. The Sage paper¹ is the primary citation for that work. Sage Plus preserves upstream history, authorship, and licensing. Downstream implementation and benchmark contributions remain attributable through this repository’s Git history. The benchmark datasets and entrapment methodology are credited in the References and the corresponding evaluation sections.

This document is project documentation with no individual author byline. It is not a peer-reviewed paper or an official upstream Sage benchmark. Results apply to the frozen software identities, inputs, and conditions described here. Reporting errors can be raised through the Sage Plus repository’s issue tracker.

The report was assembled from source code, implementation records, and benchmark summaries with assistance from OpenAI Codex. Its build tooling derives from `pgarrett-scripps/paper-scaffold`, with the scaffold license retained alongside the report source.

References

1. Lazear, M. R. Sage: An Open-Source Tool for Fast Proteomics Searching and Quantification at Scale. *Journal of Proteome Research* **22**, 3652–3659 (2023).
2. Chick, J. M. *et al.* A Mass-Tolerant Database Search Identifies a Large Proportion of Unassigned Spectra in Shotgun Proteomics as Modified Peptides. *Nature Biotechnology* **33**, 743–749 (2015).
3. Van Puyvelde, B. *et al.* A Comprehensive LFQ Benchmark Dataset on Modern Day Acquisition Strategies in Proteomics. *Scientific Data* **9**, 126 (2022).
4. Wen, B. *et al.* Assessment of False Discovery Rate Control in Tandem Mass Spectrometry Analysis Using Entrapment. *Nature Methods* **22**, 1454–1463 (2025).
5. Marx, H. *et al.* A large synthetic peptide and phosphopeptide reference library for mass spectrometry-based proteomics. *Nature Biotechnology* **31**, 557–564 (2013).

Supporting Information

Sage Plus Development and Benchmark Report

This Supporting Information records the inputs, software identities, and additional paired results for the Sage and Sage Plus versions named in the main text.

S1 Frozen software and evidence

The software pair is Sage `v0.15.0-beta.2` and Sage Plus `v0.1.0-beta.3`. The tag names belong to different projects. Version identity is fixed for this report, not updated automatically when a repository changes. Release metadata was checked on September 15, 2026.

The Sage source commit is:

`df9219951cc9a54cf4cd55d76541af24b687bd3d`

The Sage executable has the following Secure Hash Algorithm 256-bit (SHA-256) digest:

`065f9c7b445d2d5f4d3f23be007f123231e135b0344a586a656666585f8c402e`

The Sage Plus release tag resolves to source commit:

`9bfc8acbc3002d83aa3f27af5f848376b5c5c9be`

Sage Plus executable digest:

`44ed3fbde2a159b9bb4cdbfa1bf70d4e50a450aeb4a1e6d228efe81b857f43f1`

FDRBench 1.1.1 uses source commit:

`3b619a9acf60d7292fb651a00da55f58cb67fb79`

The frozen summary is `benchmarks/scientific-results/20260914/pilot-summary.json`. Its environment block records executable hashes and the host. The adjacent `FINALIZATION.json`, `evidence-audit.json`, and `archive-verification.json` record the evidence checks. Detailed results, reference receipts, conversion records, and frozen plans remain under `/data/sage-plus-scientific/20260914`.

The local archive was reopened and 3,981 members were individually verified. Large original and converted spectra remain external to that archive, with their identities recorded in receipts and manifests. A local archive is not an external deposition. Rebuilding tables from the committed summary is distinct from reacquiring inputs and repeating the searches. Original failures and amended configurations remain separately identifiable.

S2 Development source map

The development inventory uses the Sage and Sage Plus source commits recorded in Section S1. The file `analysis/data/development-changes.json` records the table text, source paths, and content digests of the inspected release files. The following paths are relative to those repository snapshots.

Input readers are in `crates/sage-cloudpath/src/`. Sage Plus adds `mzmlb.rs` and `thermoraw.rs`. Peptide and cleavage input changes are in `crates/sage/src/database.rs` and `cleavage.rs`.

Canonical result schemas and writers are in `crates/sage-cloudpath/src/parquet/`, including `results.rs`, `lfq.rs`, and `sites.rs`.

Peptide storage is defined in `crates/sage/src/peptide.rs` and `sequence.rs`. The fragment index is in `database.rs`, and spectrum storage and isotope processing are in `spectrum.rs`. Resource estimation and limits are in `crates/sage-cli/src/memory.rs` and the runner modules. API events and summaries are defined by `api.rs`, `events.rs`, and `runner/artifacts.rs`. Persistent worker orchestration is in `crates/sage-mcp/src/lib.rs`.

Scientific extensions are implemented in `crates/sage/src/mass_calibration.rs`, `scoring.rs`, and the retention and mobility modules under `ml/`. Modification semantics and neutral losses are defined in `modification.rs` and `ion_series.rs`. Quantification, localization, and empirical library export are in `lfq.rs`, `ptm.rs`, `ambiguity.rs`, and `spectral_library.rs`. The released `CHANGELOG.md`, `DOCS.md`, and `UPSTREAM.md` supply the corresponding interface and maintenance descriptions.

This map establishes the provenance of the implementation description. Benchmark provenance remains tied to the frozen executable hashes. A documented capability and a measured endpoint are distinguished in Section 2.

S3 Public input order and reference amendment

The HEK file indices are defined by the following deposited mzXML basenames. Their derived MGF files preserve those names.

- `PXD001468-0: b1906_293T_proteinID_01A_QE3_122212`.
- `PXD001468-1: b1937_293T_proteinID_01B_QE3_122212`.

The mixture indices follow the order in the frozen comparison plan. All use the `LFQ_Orbitrap_DDA_` prefix and the first injection.

- `PXD028735-0: Condition_A_Sample_Alpha_01`.
- `PXD028735-1: Condition_B_Sample_Alpha_01`.
- `PXD028735-2: Condition_A_Sample_Beta_01`.
- `PXD028735-3: Condition_B_Sample_Beta_01`.

Repeat timing used the index-zero file of each study. The acquisition receipts provide complete source filenames and URLs. Public identification pairs are indexed in the summary, so the association can be checked without interpreting figure order.

The original mixture reference contained undefined residues in these *E. coli* entries:

`P33369 P45766 P75901 P39901 P76000 P37003 P58095`

The amended `hye-irt-defined.fasta` excluded each listed entry for both engines. It did not replace unknown residues with invented amino acids or retain the known subsequences of excluded proteins. The original comparison plan and failed Sage Plus outputs were retained alongside the amended `public-comparison-v2` results. Unamended Sage results were not substituted for the matched amended-reference comparison.

The community sample annotation named a different yeast species from the primary methods. The reagent identified in those methods was Promega V7461. Its manufacturer documentation identified *S. cerevisiae*, which determined the pilot's reference choice. Manufacturer technical

manual TM410 records that reagent identity. Reference completeness and the precise *E. coli* strain remain limitations rather than assumptions silently resolved by the amendment.

S4 Additional paired resource measurements

Table S1 summarizes the entrapment workloads. Each engine has one search per selected file and construction seed. These medians combine different inputs, so they are descriptive workload summaries rather than repeat timing estimates.

Study	Engine	Searches	Median seconds	Median peak MiB
HEK	Sage	6	19.30	3866.4
HEK	Sage Plus	6	22.98	3148.8
Mixture	Sage	6	40.83	5330.8
Mixture	Sage Plus	6	48.39	4337.7

Table S1: Resource use with entrapment-expanded references. Each engine searched two selected files with three construction seeds per study. Medians span those file-by-seed searches. The mixture reference includes the documented exclusions. MiB describes peak resident memory, not the process address-space limit.

The local HEK conventional and common-modification results appear in Table S2. These searches used the same frozen release pair as the main text. Acquisition could overlap these runs, so their wall times are not part of the controlled public timing endpoint. The common-modification workload adds methionine oxidation and peptide N-terminal acetylation under the shared search configuration. The local biological input has unverified redistribution provenance and is not offered as a public replication dataset.

Workload	Engine	Seconds	Peak MiB	Target PSMs
standard	Sage	5.71	1903.9	2214
standard	Sage Plus	6.41	1502.6	2203
common-mods	Sage	12.47	5348.5	2266
common-mods	Sage Plus	15.06	3633.5	2270

Table S2: Contextual local HEK comparison of the specified Sage and Sage Plus releases. Values are medians of three measured trials after one warmup. Target PSMs pass the reported one-percent spectrum q-value threshold. Possible overlap with acquisition activity limits timing interpretation.

The broad PTM workload did not complete with either engine under the selected limits. Plus applied its additional modified-database guard and Sage encountered allocation failure under the shared address-space ceiling. Failed warmups and attempts are retained. Their elapsed time is not summarized as the time required to finish the search.

S5 Threshold-dependent entrapment results

Table S3 retains every prespecified nominal threshold from the independent audit. Target counts and FDP estimates are averaged across the same file-by-seed combinations. A higher target count at a nominal q-value is not a matched-error sensitivity result.

Study	Engine	Nominal q (%)	Mean targets	Mean paired FDP (%)
HEK	Sage	0.1	8636.0	0.071
HEK	Sage	0.5	9822.5	0.492
HEK	Sage	1.0	10046.3	1.017
HEK	Sage	2.0	10229.5	2.095
HEK	Sage	5.0	10515.5	5.113
HEK	Sage Plus	0.1	8739.0	0.096
HEK	Sage Plus	0.5	9833.7	0.515
HEK	Sage Plus	1.0	10067.7	0.998
HEK	Sage Plus	2.0	10249.2	2.145
HEK	Sage Plus	5.0	10540.0	5.235
Mixture	Sage	0.1	30245.0	0.091
Mixture	Sage	0.5	35214.3	0.533
Mixture	Sage	1.0	37165.7	1.029
Mixture	Sage	2.0	38893.8	2.034
Mixture	Sage	5.0	40856.2	5.018
Mixture	Sage Plus	0.1	30293.0	0.096
Mixture	Sage Plus	0.5	35339.2	0.545
Mixture	Sage Plus	1.0	37291.3	1.057
Mixture	Sage Plus	2.0	39003.7	2.084
Mixture	Sage Plus	5.0	40952.3	5.039

Table S3: Threshold-specific peptide entrapment results. Each row averages two files and three shared construction seeds for the named engine and study. FDP uses the conservative paired bound over exact target-partner score ties. Nominal q-values and independently estimated FDP are reported in percent.

Construction seeds were 20260914, 20260915, and 20260916. The bootstrap used seed 20260914 and 10,000 resamples. It resampled the file and construction-seed dimensions while preserving engine pairing. The bounds in the main text use the 2.5th and 97.5th percentiles. These are conditional descriptive intervals over the selected files and seeds.

The audit checked counts against official FDRBench output and explicitly considered exact score ties. Empty discovery denominators remain undefined. This convention prevents absent evidence from being presented as a zero-error success. Search failures are also retained as failures rather than converted to empty discovery sets.

S6 Public identification counts and worker scaling

Table S4 gives the absolute counts underlying the main identification figure. Peptidoform counts use peptide q-values and PSM counts use spectrum q-values. These filters are distinct. Decoy PSM counts document the retained confidence-filtered output, but they are not substituted for the independent entrapment estimator.

Input	Engine	PSMs	Peptidoforms	Decoy PSMs
PXD001468-0	Sage	16,866	9,264	167
PXD001468-0	Sage Plus	16,917	9,293	168
PXD001468-1	Sage	19,406	11,084	193
PXD001468-1	Sage Plus	19,471	11,103	193
PXD028735-0	Sage	55,457	39,008	553
PXD028735-0	Sage Plus	55,612	39,117	555
PXD028735-1	Sage	55,940	38,925	558
PXD028735-1	Sage Plus	56,044	38,982	559
PXD028735-2	Sage	58,337	40,047	582
PXD028735-2	Sage Plus	58,508	40,117	584
PXD028735-3	Sage	57,174	39,147	570
PXD028735-3	Sage Plus	57,321	39,271	572

Table S4: Per-file accepted identifications at the one-percent threshold of the indicated confidence level. The common normalizer reads underlying result tables from both engines.

Input indices follow Section S3. Peptidoforms are counted separately from spectra.

Table S5 reports every engine and worker-count summary from the extension. Each range spans three measured repeats following one warmup. The PSM range is shown alongside time so that a repeat with changed yield cannot be hidden by a single aggregate timing value. Assignment differences are analyzed separately in the main text.

Engine	Workers	Seconds	Time range	Peak MiB	PSM range
Sage	1	39.17	38.90 to 39.53	1752.6	16,866 to 16,866
Sage Plus	1	39.84	39.01 to 39.95	1423.7	16,917 to 16,917
Sage	2	22.22	21.83 to 22.78	1791.7	16,866 to 16,866
Sage Plus	2	23.18	22.65 to 24.88	1424.2	16,917 to 16,917
Sage	4	14.48	13.92 to 14.80	2096.2	16,866 to 16,866
Sage Plus	4	14.07	13.99 to 14.20	1424.7	16,917 to 16,917
Sage	8	10.42	10.26 to 10.54	2106.4	16,866 to 16,866
Sage Plus	8	9.82	9.75 to 9.97	1500.7	16,917 to 16,917

Table S5: Matched worker-scaling summaries from the report extension. Seconds and peak resident MiB are medians. Time and PSM ranges span measured repeats. Warmups remain in the execution records but are excluded from these summaries.

S7 Quantitative comparison denominators

Table S6 restricts the accuracy comparison to peptide and preparation pairs accepted by both releases. Combined-charge precursors use the same empty charge key in both formats. Numeric modification spelling is normalized without changing mass values or modification positions. The shared set is intersected at the ratio-pair level, so both conditions have positive accepted intensity in both engines.

Species	Shared pairs	Sage abs. error	Plus abs. error	Median ratio difference
ecoli	1,788	0.448	0.447	0.007
human	33,792	0.237	0.235	0.007
yeast	10,183	0.273	0.271	0.006

Table S6: Ratio accuracy on identical accepted peptide-preparation pairs. Absolute error is measured against the expected species log-base-two B/A ratio. The last column is the median absolute difference between the two engines’ observed ratios on the same pairs. These metrics assess common features and complement the engine-specific sets in the main text.

The control denominator in Table S7 includes positive finite accepted precursor-file intensities with an unambiguous species assignment. A foreign sequence also occurring in the human reference after isoleucine/leucine normalization is excluded. Retention-time standards and unmapped proteins are outside the species-control denominator. The numerator is a subset of the denominator under the same rule. The absence of direct evidence is computed from rank-one PSM results at the joint spectrum and peptide threshold. It does not use an exported confirmation flag whose meaning may differ between releases.

Engine	Control rows	Foreign	No direct MS2	Foreign, no direct MS2
Sage	24,262	5,965	8,985	5,928
Sage Plus	23,771	5,872	8,765	5,836

Table S7: Exact human-only control denominators at the primary LFQ threshold. Foreign means a yeast or E. coli assignment without a human-reference equivalent. Direct MS2 requires matching peptide and file with both spectrum and peptide q-values at most one percent. No direct MS2 is a subset of the complete control set, not an additional independent sample.

The primary LFQ threshold applies to a precursor-level score shared across files. A positive intensity in a file is therefore not accompanied by an independently calibrated recipient-file transfer probability in this analysis. The human-only control tests the intended absent-species condition. The main Limitations section describes the boundaries of its error interpretation.

S8 Synthetic-site diagnostic retained separately

Table S8 records the released Sage Plus site output after spectrum and localization filtering, deliberately before peptide filtering. All primary jointly accepted site sets were empty. The diagnostic rows therefore explain what was withheld by the primary confidence requirement. They are not an alternative validated result obtained by relaxing that requirement.

HCD input	Consistent sites	Inconsistent sites	Inconsistent (%)	Jointly accepted sites
1	1172	118	9.15	0
2	880	15	1.68	0

Table S8: Secondary Sage Plus synthesis-consistency diagnostic from the same released executable. Diagnostic events pass spectrum and localization q-values at one percent but omit the peptide filter. The final column restores the primary joint requirement. Consistency is assessed against pooled unambiguous synthesis truth, with file-to-library mapping unaudited.

The synthesis reference and truth were prepared before the original searches. The original all-site configurations are retained for both releases. Oracle site-prior runs used information from synthesis truth and are excluded from comparative accuracy claims. Development repairs after

the released snapshot are also excluded. A result from a modified executable must be assigned its own software identity and independently re-evaluated before entering a later report.

S9 Report extension and analysis provenance

The extension plan is `paper/analysis/data/report-extension/plan.json`. Its execution directory is `/data/sage-plus-scientific/report-extension-20260915`. The frozen runner records executable and input hashes, the host, commands, process limits, and output identities. Its matrix-status file retains the completion state of every planned job. The LFQ extension has one search per engine and does not support a repeated runtime estimate. Worker scaling has one warmup and three measured trials per engine and worker count.

All 34 planned extension jobs completed. The closing audit verified 314 source and output file identities.

The new analysis does not modify the original pilot summary or its analysis scripts. `collect_report.py` reads the retained raw results, normalizes the two output representations, and records SHA-256 identities of the source files in separate report snapshots. Public disagreement classifications are checked against the frozen engine-only counts. Quantitative ratio counts are checked against the common scientific metric function. The shared-set calculation uses identical peptide, charge, and preparation keys across releases.

The additional analyses were motivated by gaps identified after inspection of the pilot. They reuse previously selected public files and are explicitly post hoc extensions. Neither their additional compute repeats nor their many precursor measurements increase the number of independent biological studies. The original pilot, extension measurements, and engine-specific diagnostics remain distinguishable in the figures and captions.

S10 Regenerating the report

The report reads the frozen pilot summary through `analysis/scripts/_scientific.py`. The statistics generator supplies computed values used in prose. Table generators read the same inputs, and the figure generators plot the same frozen evidence and new report snapshots. Each asset records the analysis code and data identities that produced it. These steps do not rerun biological searches or modify the finalized evidence.

From the repository root, rebuild the report with:

```
cd paper
just assets
just fmt
just docx
just paper
just verify
just check-stats-deep
```

To repeat the biological analysis, first inspect `benchmarks/SCIENTIFIC_PILOT.md` and the retained acquisition, conversion, reference, and search plans. Supply the pinned executables and verified inputs at the paths expected by a copied plan. Use a new output directory for a new experiment. Do not overwrite the frozen evidence or describe a changed binary as the originally evaluated release.

The small report snapshots can be regenerated from the retained raw results:

```
cd paper/analysis
uv run scripts/collect_report.py public
uv run scripts/collect_report.py ptm
uv run scripts/collect_report.py lfq
uv run scripts/collect_report.py scaling
```

These commands derive tables and distributions from existing searches. Repeating an executable benchmark is a different operation that requires the pinned binary, complete input files, and a newly named output directory. The report's figures can be rebuilt from the snapshots without copying the large spectra into the report directory.